6CCSACCM: Cooperation, Coordination and Multi-Agent Systems

Week 1

Non-cooperative games (part I)

Maria Polukarov

The Ice Cream Wars

- Ed and Ted are selling identical ice cream bars on a beach, which we model as the interval [0,1].
- Since the ice cream bars are identical, customers always go to the vendor closest to them.
- Initially, Ed sets up his cart at the $1/4$ mark on the beach and Ted sets up his cart at the $3/4$ mark, so both vendors got $1/2$ of the customers.
- After some time, Ted employs a useful deviation from his original strategy and moves to the $1/2$ mark on the beach, and Ted now gets $5/8$ of the customers while Ed only gets $3/8$ of the customers.

The Ice Cream Wars

- We have $N = \{1, 2\}$ and $S_1 = S_2 = [0, 1]$, where a strategy $s_i \in S_i$ represents setting up a cart at s_i on the beach.
- Further, we have that

$$U_i(s_i, s_{-i}) = \begin{cases} \frac{s_i + s_{-i}}{2} & \text{if } s_i < s_{-i} \\ 1 - \frac{s_i + s_{-i}}{2} & \text{if } s_i > s_{-i} \\ \frac{1}{2} & \text{if } s_i = s_{-i} \end{cases}$$

Working on a joint project

- 2 students work on a joint project.
- Each either works hard or goofs off.

	Work hard	Goof off
Work hard	$2, 2$	$0, 3$
Goof off	$3, 0$	$1, 1$

Duopoly

- 2 firms produce the same good.
- Each charges either a high price or a low price.

	High	Low
High	$1000, 1000$	$-200, 1200$
Low	$1200, -200$	$600, 600$

Prisoner's dilemma: general form

	C	D
C	R, R	S, T
D	T, S	P, P

- $T > R > P > S$

Tragedy of the Commons

- A social dilemma where individuals have an incentive to overconsume a common resource and act in their own self-interest at the expense of society.

BEAUTY CONTEST!

"Beauty contest"

- Each participant i chooses an integer from $S_i = \{0, 1, \ldots, 100\}$.
- Player i whose strategy $s_i \in S_i$ is *closest to 2/3 of the average of all strategies*, wins the contest!
- That is, each player's aim is to minimise the difference

$$\left| s_i - \frac{2}{3} \frac{\sum_{j \in \mathbf{N}} s_j}{N} \right|$$

The Professor's Dilemma

- This is a game played between a professor and the class.

PAYOFFS	**Listen**	**Sleep**
Make effort	$(10^6, 10^6)$	$(-10, 0)$
Slack off	$(0, -10)$	$(0, 0)$

- No player has a dominant or dominated strategy.
- The best strategy for a player depends on what the other player is doing.

The Ice Cream Wars

- We have $N = \{1, 2\}$ and $S_1 = S_2 = [0, 1]$, where a strategy $s_i \in S_i$ represents setting up a cart at s_i on the beach.
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$$U_i(s_i, s_{-i}) = \begin{cases} \frac{s_i + s_{-i}}{2} & \text{if } s_i < s_{-i} \\ 1 - \frac{s_i + s_{-i}}{2} & \text{if } s_i > s_{-i} \\ \frac{1}{2} & \text{if } s_i = s_{-i} \end{cases}$$

Battle of Sexes

- Battle of Sexes is another generalisation of a coordination game where both players want to choose the same action, but have different preferences over possible consistent outcomes.

PAYOFFS	**Theater**	**Football**
Theater	$(2, 1)$	$(0, 0)$
Football	$(0, 0)$	$(1, 2)$

- (**Theater**, **Theater**) and (**Football**, **Football**) are both NE.

Exercise 1: removal dominated strategies

- Solve the game by eliminating dominated strategies.
- Verify that the resulting solution is a Nash equilibrium.

	L	C	R
T	$73, 25$	$57, 42$	$66, 32$
M	$80, 26$	$35, 12$	$32, 54$
B	$28, 27$	$63, 31$	$54, 29$

Exercise 1

- R dominates L:

	C	R
T	$57, 42$	$66, 32$
M	$35, 12$	$32, 54$
B	$63, 31$	$54, 29$

Exercise 1

- T and B dominate M:

	C	R
T	$57, 42$	$66, 32$
B	$63, 31$	$54, 29$

Exercise 1

- C dominates R:

	C
T	$57, 42$
B	$63, 31$

Exercise 1

- B dominates T:

	C
B	$63, 31$

Exercise 1

- (B, C) is a Nash equilibrium: each player is playing their best response to the other player's strategy.

	L	C	R
T	$73, 25$	$57, 42$	$66, 32$
M	$80, 26$	$35, 12$	$32, 54$
B	$28, 27$	$\mathbf{63, 31}$	$54, 29$

Exercise 2: best response

- Find the players' best responses to each other's strategies.
- Find all (pure strategy) Nash equilibria.

	L	C	R
T	$2, 2$	$1, 3$	$0, 1$
M	$3, 1$	$0, 0$	$0, 0$
B	$1, 0$	$0, 0$	$0, 0$

Exercise 2

- Best responses:

	L	C	R
T	$2, 2$	$1^*, 3^*$	$0^*, 1$
M	$3^*, 1^*$	$0, 0$	$0^*, 0$
B	$1, 0^*$	$0, 0^*$	$0^*, 0^*$

Exercise 2

- Nash equilibria:

	L	C	R
T	$2, 2$	$\mathbf{1}, \mathbf{3}$	$0, 1$
M	$\mathbf{3}, \mathbf{1}$	$0, 0$	$0, 0$
B	$1, 0$	$0, 0$	$\mathbf{0}, \mathbf{0}$

Exercise 3: dividing money

Determine each player's best responses and find Nash equilibria of the following game:

- Two people have £10 to divide between themselves.
- Each player i names a number $x_i \in \{0, 1, \ldots, 10\}$.
- If $x_1 + x_2 \leq 10$ then each player get his claim (and the remainder is destroyed).
- If $x_1 + x_2 > 10$ **and** $x_1 \neq x_2$ then $i \in \arg\min\{x_1, x_2\}$ gets x_i and the other gets the rest.
- If $x_1 + x_2 > 10$ **and** $x_1 = x_2$ then each gets £5.

Exercise 3

Best responses:

Other player's action	Sets of best responses
0	$\{10\}$
1	$\{9, 10\}$
2	$\{8, 9, 10\}$
3	$\{7, 8, 9, 10\}$
4	$\{6, 7, 8, 9, 10\}$
5	$\{5, 6, 7, 8, 9, 10\}$
6	$\{5, 6\}$
7	$\{6\}$
8	$\{7\}$
9	$\{8\}$
10	$\{9\}$

Exercise 3

Nash equilibria:

Other player's action	Sets of best responses
0	$\{10\}$
1	$\{9, 10\}$
2	$\{8, 9, 10\}$
3	$\{7, 8, 9, 10\}$
4	$\{6, 7, 8, 9, 10\}$
5	$\{\mathbf{5}, \mathbf{6}, 7, 8, 9, 10\}$
6	$\{\mathbf{5}, \mathbf{6}\}$
7	$\{6\}$
8	$\{7\}$
9	$\{8\}$
10	$\{9\}$

Battle of Sexes

PAYOFFS	**Theater**	**Football**
Theater	$(2, 1)$	$(0, 0)$
Football	$(0, 0)$	$(1, 2)$

Battle of Sexes

PAYOFFS	**Theater**	**Football**
Theater	$(2, 1)$	$(0, 0)$
Football	$(0, 0)$	$(1, 3)$

THANK YOU!